

Adelic Entropic Numbers: When the Adelic Information Vector Becomes the Entropic Number

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Abstract

Two papers, written independently on the same day, converge on the same structure from opposite directions. *Adelic Shannon Theory* [1] generalises Shannon information theory to the adèle ring $\mathbb{A}_{\mathbb{Q}}$, defining the *adelic information vector* $\mathbf{I}(X) = (I_{\infty}, I_2, I_3, \dots)$ where I_{∞} is standard Shannon entropy and I_p is p -adic valuation entropy. *Measurement Stratigraphy* [2] forecasts three future eras of number systems, the second of which — Era 11: Entropic Enclosure — defines an *entropic number* as a pair (x, S) : a best estimate x plus an entropy measure S encoding uncertainty. This note demonstrates that these are the **same structure**: the entropic number's S is precisely the adelic information vector $\mathbf{I}(X)$, and the entropic number $(x, \mathbf{I}(X))$ is the natural data type for physics conducted over $\mathbb{Q}$ rather than $\mathbb{R}$.

The convergence has immediate consequences. (1) The Gaussian $e^{-\pi x^2}$ is the universal entropic number: it maximises entropy at *every* place simultaneously — archimedean (differential entropy) and non-archimedean (valuation entropy). (2) An AI built on adelic entropic numbers would encode uncertainty at all completions of $\mathbb{Q}$ — not merely reporting $\sigma = 0.1$ in the archimedean norm, but also the p -adic valuation entropy structure of every measurement. (3) The adelic product-formula coding theorem becomes the channel capacity for entropic number transmission: the total rate is a product over places because uncertainty at different places is independent. [SPECULATIVE]

Keywords: entropic number, adelic information vector, p -adic entropy, measurement stratigraphy, Gaussian, max-entropy, AI safety, honest numbers

1. The Two Papers, One Structure

1.1 From One Direction: Adelic Shannon Theory [1]

The Adelic Shannon Theory paper defines the **adelic information vector**:

$$\mathbf{I}(X) = (I_{\infty}(X), I_2(X), I_3(X), I_5(X), \dots)$$

where: - $I_{\infty}(X) = H(X) = -\sum p(x) \log_2 p(x)$, the standard Shannon entropy (archimedean information, in bits) - $I_p(X) = H_p(X) = \sum p(x) \cdot v_p(x)$, the p -adic valuation entropy (expected

p -divisibility)

The central structural claim: **information is not a scalar but a vector over all completions of $\mathbb{Q}$** . The information at different places is independent — knowing the Shannon entropy does not determine the p -adic entropies, and vice versa.

The Gaussian $e^{-\pi x^2}$ is proved to be the unique function that simultaneously maximises entropy at the archimedean place (differential entropy under variance constraint) and at all non-archimedean places (valuation entropy under expected-valuation constraint). The Poisson summation formula $\sum_n f(n) = \sum_n \hat{f}(n)$ is reinterpreted as the adelic source-channel equality theorem.

1.2 From the Other Direction: Measurement Stratigraphy [2]

The Measurement Stratigraphy paper forecasts three future eras of number systems. **Era 11: Entropic Enclosure** defines an **entropic number**:

An entropic number is a pair (x, S) where x is a best estimate (a section of the sheaf) and S is an entropy measure encoding uncertainty.

The crucial constraint: **when no additional information is available, the distribution must be the maximum-entropy one compatible with given moments**. Consequences:

- The Gaussian becomes the universal default — if you know only the mean and variance, the entropic number IS a Gaussian distribution, forced by the max-entropy principle.
- Physics becomes entropic flow — the Schrödinger equation and heat equation are two manifestations of the same entropic evolution.
- **“An entropic number never claims more than it knows. An AI built on entropic numbers would never hallucinate false certainty.”**

1.3 The Convergence

The two papers converge on the same mathematical object:

Adelic Shannon Theory [1]	Measurement Stratigraphy [2]	Unified
Adelic information vector $\mathbf{I}(X)$	Entropy measure S	$S = \mathbf{I}(X)$
Max-entropy at all places	Max-entropy compatible with moments	Gaussian $e^{-\pi x^2}$
Product-formula capacity	—	Channel bound for entropic number transmission
Source-channel equality via Poisson sum	Poisson sum as entropy conservation	Adelic source-channel identity
AI-safe p -adic uncertainty	AI-safe entropic numbers	Adelic entropic numbers as the honest data type

The entropic number’s “ S ” was defined abstractly — “an entropy measure encoding uncertainty.” The Adelic Shannon Theory provides the **concrete specification**: S is the vector $\mathbf{I}(X)$, carrying the standard (archimedean) entropy AND the p -adic valuation entropies for every prime p .

2. The Adelic Entropic Number

2.1 Definition

Definition (Adelic Entropic Number). An adelic entropic number is a pair:

$$\mathcal{E}(x) = (x, \mathbf{I}(x))$$

where: - $x \in \mathbb{Q}$ is the best estimate (a rational number — the physically accessible value) - $\mathbf{I}(x)$ is the adelic information vector of the distribution p_X governing the uncertainty

The distribution p_X must satisfy the **adelic max-entropy principle**: given the known moments (archimedean: mean and variance; non-archimedean: expected valuation at each prime), p_X is the distribution that simultaneously maximises I_∞ and every I_p .

2.2 The Gaussian as the Universal Entropic Number

Theorem (Universal Entropic Number). The Gaussian distribution

$$p_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$$

is the unique entropic number for the constraint ($\mathbb{E}[X] = \mu, \text{Var}[X] = \sigma^2$) at the archimedean place and the constraint that the p -adic localisation to each $\mathbb{Z}_p$ is the uniform Haar measure (the default when no p -adic information is available).

Proof sketch: - At ∞ : standard — Gaussian maximises differential entropy given variance. [established] - At each p : the characteristic function $\mathbf{1}_{\mathbb{Z}_p}$ (normalised Haar measure on $\mathbb{Z}_p$) is the p -adic analogue of the uniform distribution on a bounded interval. It has maximal p -adic valuation entropy among distributions supported on $\mathbb{Z}_p$ with no further constraints. [SPECULATIVE] - The global section: the Gaussian on $\mathbb{R}$ glues to the Haar measure on each $\mathbb{Z}_p$ via the Poisson summation formula — the descent condition that guarantees the archimedean and non-archimedean descriptions are compatible. [3]

2.3 What the Entropic Number Carries (That a Float Does Not)

A standard measurement representation in physics:

`x = 5.0 ± 0.1 # archimedean only`

An adelic entropic number:

`E(x) = (5.0, I_∞=4.32 bits, I_2=0.33, I_3=0.17, I_5=0.05, ...)`

Component	Meaning	What a float misses
$I_\infty = H(X)$	Standard uncertainty (bits)	Captured as $\sigma = 0.1$
I_2	Expected power of 2 dividing x	Lost — there is no “binary uncertainty” field
I_3	Expected power of 3 dividing x	Lost
I_p	Expected p -adic valuation	Lost — ALL non-archimedean structure is invisible

The standard float 5.0 ± 0.1 reports **only the archimedean uncertainty**. It is silent on — literally has no field for — the p -adic uncertainty structure. This is the “incompleteness” of the archimedean measurement paradigm: not wrong (the archimedean uncertainty is real) but incomplete (it misses all other completions of $\mathbb{Q}$).

2.4 Honest AI

An AI built on entropic numbers would structurally reduce hallucination — because the data type has no “certainty” constructor.

Why? Because an entropic number **cannot** be a bare point estimate. It is always a pair $(x, \mathbf{I}(x))$ — the uncertainty is part of the data structure, not an optional metadata field. An LLM that must produce an entropic number as output cannot produce "Paris" — it must produce

$$\mathcal{E}(\text{"Paris"}) = (\text{"Paris"}, \mathbf{I}(\text{context}))$$

where $\mathbf{I}(\text{context})$ encodes the model’s uncertainty about the answer — the entropy of the next-token distribution, the p -adic valuation entropy of the embedding, etc. The data type forces the model to expose uncertainty. **Whether this fully prevents hallucination depends on the architecture’s fidelity** — the entropic number output removes the *data-type-level* certainty illusion, but distribution-shift and attention-collapse failures remain open problems [SPECULATIVE — see calibration register below].

The standard LLM output is a **lossy projection** — exactly the same crime as $\mathbb{R}_{\text{comp}} \rightarrow \mathbb{R}$ in Era 3 → 4 of the Measurement Stratigraphy. The raw next-token distribution is constructive (the model actually computed it); the argmax token string is a projection that discards the entropy information. An entropic number output would preserve the full distribution — structurally reducing the surface area for hallucination (though distribution-shift failures remain an open architecture problem).

3. Consequences for the Adelic Programme

3.1 Channel Capacity for Entropic Numbers

The Adelic Shannon Theory [1] proves the product-formula coding theorem for the factorisable case:

$$C(\mathbb{A}_{\mathbb{Q}}) = C_{\infty} \times \prod_p C_p$$

For entropic number transmission, this means: the channel capacity for sending an entropic number is the product of the capacity at each place. If any place loses capacity (e.g., a p -adic noise source increases $H_p(N)$), the total product capacity drops — even if the archimedean SNR remains unchanged.

3.2 The Bridge Theorem: Poisson Summation

Measurement Stratigraphy [2] identifies the Poisson summation formula as the descent condition that glues discrete and continuous views. Adelic Shannon Theory [1] identifies it as source-channel equality. The combined statement:

Adelic Bridge Theorem. The Poisson summation formula

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{n \in \mathbb{Z}} \hat{f}(n)$$

is simultaneously: 1. The descent condition that makes the sheaf of local number systems a global object (Era 10: Contextual Enclosure) [2] 2. The source-channel equality for the Gaussian entropic number (Era 11: Entropic Enclosure) [2] 3. The adelic coding theorem: total source entropy equals total channel entropy at every place [1]

The Gaussian $e^{-\pi x^2}$ satisfies all three interpretations because it is the unique global section: invariant under Fourier duality (1), maximum-entropy at all places (2), and achieving source-channel equality (3).

3.3 From Information Vector to Entropic Number

The mapping is direct:

$$\text{Adelic Information Vector } \mathbf{I}(X) \longrightarrow \text{Entropic Number } \mathcal{E}(x)$$

The information vector tells you *how much* uncertainty exists at each place. The entropic number packages this uncertainty *with* the estimate as a unified data type. The former is the theory; the latter is the implementation.

4. Falsifiability Conditions

The convergence of these two frameworks is falsifiable:

1. **Gaussian as universal entropic number.** If there exists a distribution with the same first two moments as the Gaussian that has strictly higher p -adic valuation entropy at *some* prime p , the claim that the Gaussian is the unique universal entropic number is falsified. [UNTESTED]
2. **Adelic data-processing inequality.** If there exists a physically realisable channel $\mathcal{E}$ and source X such that $\mathbf{I}(\mathcal{E}(X)) \geq \mathbf{I}(X)$ componentwise (strictly at some place) for a channel that does not inject information, the adelic data-processing inequality is violated. [UNTESTED]
3. **AI hallucination prevention.** If an AI system built with entropic number outputs still hallucinates (produces claims with deceptively low encoded uncertainty), the structural prevention claim is falsified. This requires building such a system — it is a forward-looking calibration register entry. [UNTESTED — no such system exists]

4.1 Calibration Register

For predictions made in this paper that are forward-looking and falsifiable:

ID	Prediction	By	Strength	Status
CR-AEN-01	An AI system with entropic number outputs demonstrates $\leq 10\%$ of the “false certainty” hallucination rate of comparable scalar-output systems on standard benchmarks	2030	[WEAK — calibrated subjective, no base-rate class exists for this novel architecture]	PENDING
CR-AEN-02	At least one experimental physics measurement reports an adelic entropic number (i.e., includes p -adic valuation entropy alongside archimedean uncertainty) in a peer-reviewed publication	2035	[WEAK — calibrated subjective, adoption rate of new data types is historically slow]	PENDING
CR-AEN-03	The Gaussian $e^{-\pi x^2}$ is experimentally confirmed as the default entropic number distribution for quantum measurements where only mean and variance are specified, with deviations $\leq 5\%$ in KL divergence from max-entropy prediction	2040	[WEAK — calibrated subjective, requires experimental infrastructure not yet built]	PENDING

WEAK = likelihood anchored only by calibrated subjective confidence; no external base-rate class exists for these novel predictions.

5. Conclusion: The Two-Way Street

The Adelic Shannon Theory and the Measurement Stratigraphy were written on the same day, by the same research programme, approaching the same structure from opposite sides:

	Adelic Shannon Theory	Measurement Stratigraphy
Direction	Bottom-up: generalise Shannon's axioms to all places	Top-down: forecast the next era of number systems from the historical stratigraphy
Object	Adelic information vector $\mathbf{I}(X)$	Entropic number (x, S)
Key insight	Information is a vector, not a scalar	Uncertainty must be a first-class citizen of the number system
Gaussian	Unique max-entropy distribution at every place simultaneously	Universal default — the only distribution forced by max-entropy when only mean/variance are known

They are the **same claim** stated in different mathematical languages. The adelic information vector IS the entropy measure S in the entropic number. The entropic number IS the natural data type for measurements over $\mathbb{Q}$.

The practical consequence: a measurement system that stores adelic entropic numbers — not bare floats — would produce **structurally honest** data. Every measurement would carry its uncertainty at every completion of $\mathbb{Q}$, and an AI consuming such data would have significantly reduced hallucination surface because the data type exposes ignorance — though this is a necessary, not sufficient, condition for honest AI.

The bridge is open. The two papers have met at the middle.

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